Multi-Structure KELM With Attention Fusion Strategy for Hyperspectral Image Classification

Le Sun, *Member, IEEE*, Yu Fang, Yuwen Chen, Wei Huang, *Member, IEEE*, Zebin Wu, *Senior Member, IEEE*, and Byeungwoo Jeon, *Senior Member, IEEE*

***Abstract*— Hyperspectral image (HSI) classification refers to accurately corresponding each pixel in an HSI to a land-cover label. Recently, the successful application of multiscale and multifeature methods has greatly improved the performance of HSI classification due to their enhanced utilization of the available spectral–spatial information. However, as the number of scales and the number of features increases, it becomes more difficult to achieve an optimal degree of fusion for multiple classifiers [e.g., kernel extreme learning machine (KELM)]. On the other hand, a limited sample size of the HSI may cause overfitting problems, which seriously affects the classification accuracy. Therefore, in this article, a novel multi-structure KELM with attention fusion strategy (MSAF-KELM) is proposed to achieve accurate fusion of multiple classifiers for effective HSI classification with ultrasmall sample rates. First, a multi-structure network is built, which combines multiple scales and multiple features to extract abundant spectral–spatial information. Second, a fast and efficient KELM is employed to enable rapid classification. Finally, a weighted self-attention fusion strategy (WSAFS) is introduced, which combines the output weights of each KELM subbranch and the self-attention mechanism to achieve an efficient fusion result on multi-structure networks. We conducted experiments on four types of HSI datasets with different evaluation methods and compared them with several classical and state-of-the-art methods, which demonstrate the excellent performance of our method on ultrasmall sample rates. The code is available at https://github.com/Fang666666/MSAF-KELM for reproducibility.**

Manuscript received 10 July 2022; revised 23 August 2022; accepted 16 September 2022. Date of publication 20 September 2022; date of current version 3 October 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 61971233 and Grant 62076137, and in part by the "Qinglan Project" of Jiangsu Universities under Grant D202062032. *(Corresponding author: Wei Huang.)*

Le Sun is with the School of Computer Science, Nanjing University of Information Science and Technology, Nanjing 210044, China, and also with the Engineering Research Center of Digital Forensics, Ministry of Education, Nanjing University of Information Science and Technology (NUIST), Nanjing 210044, China (e-mail: sunlecncom@163.com).

Yu Fang is with the School of Computer Science, Nanjing University of Information Science and Technology, Nanjing 210044, China (e-mail: cs_yfang@nuist.edu.cn).

Yuwen Chen is with the Chongqing Institute of Green and Intelligent Technology, Chinese Academy of Sciences (CAS), Chongqing 400714, China (e-mail: chenyuwen@cigit.ac.cn).

Wei Huang is with the School of Computer and Communication Engineering, Zhengzhou University of Light Industry, Zhengzhou 450000, China (e-mail: hnhw235@163.com).

Zebin Wu is with the School of Computer Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: zebin.wu@gmail.com).

Byeungwoo Jeon is with the School of Electrical Engineering, Sungkyunkwan University, Suwon 440746, South Korea (e-mail: bjeon@skku.edu).

Digital Object Identifier 10.1109/TGRS.2022.3208165

***Index Terms*— Attention mechanisms, hyperspectral image (HSI) classification, kernel extreme learning machine (KELM), multifeature, multiscale (MS).**

I. Introduction

HYPERSPECTRAL images (HSIs) have received more attention recently because they contain abundant spatial information and rich spectral bands [1]. Compared with other remote sensing images, the HSI stores more information in the form of cubic data [2]. Due to the great value of the information process and analysis for HSIs, researchers have carried out a series of technical processing techniques [3], [4], [5], [6], [7], [8], [9], [10], such as hyperspectral classification. HSI has been widely utilized in many fields, including mineral resource development [11], precision agriculture [12], and medical imaging technology [13], which all rely on HSI classification.

In the past few decades, many researchers have proposed numerous traditional machine learning methods for HSI classification, such as the random forest (RF) [14], the K-nearest neighbor algorithm [15], the Bayesian model [16], and the support vector machine (SVM) [17]. However, these traditional methods insufficiently utilize spectral–spatial features to establish the relationship between pixels in the spatial dimension [18]. The advancement of various feature extraction algorithms and their successful application in HSI classification have greatly improved the classification accuracy [19]. The most representative handcrafted extraction features are mathematical morphological features, such as morphological profile (MP) [20], extended MP (EMP) [21], and extended multiattribute profile (EMAP) [22]. In addition, feature extraction algorithms based on subspace learning [3], such as sparse and low-rank representations [23], have contributed to the progress of classification. Moreover, subspace-based [24], discriminant analysis [25], and multiview learning [26] methods can greatly enhance the feature representation.

To further exploit the rich spectral–spatial information in HSIs and ease the limitations on the spatial representation capability, deep learning methods have been developed for HSI classification in recent years. Compared with traditional methods, deep learning methods use more advanced feature extraction mechanisms in end-to-end networks, which greatly contributes to the improvement of HSI classification accuracy. For instance, Yan *et al.* [27] input the dimensionality-reduced

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:54:29 UTC from IEEE Xplore. Restrictions apply.

images using principal component analysis (PCA) into a stacked autoencoder network for HSI classification. They used convolutional neural networks (CNNs) to more effectively exploit the spectral–spatial features in HSI. Li *et al.* [28] proposed a CNN based on pixel-pair features (CNN-PPFs) by modifying and relabeling the available samples, which greatly increased the number of available training samples. Roy *et al.* [29] combined the characteristics of 3-D and 2-D CNNs and proposed a hierarchical network structure called HybridSN. In addition, some of the latest deep learning methods have achieved encouraging results in the domain of HSI classification over recent years. Hong *et al.* [30] proposed a minibatch graph convolutional network (miniGCN), which allowed the training of large-scale graph convolutional network (GCN) with small batches, effectively improving the ability to model relationships between complex samples. Wan *et al.* [31] proposed a dual-interactive GCN that captured spatial information at different scales, further refined the modeling representation capability of the graph, and accelerated the computational speed of the convolution operation. Hong *et al.* [32], [33] and Wu *et al.* [34] even creatively developed a general multimodal deep learning network, which broke through the limitation of information diversity and greatly improved the performance of remote sensing image classification in complex scenes. Although deep learning methods have achieved great success in the field of HSI classification, these methods generally have deep network layers and require a large number of training samples when processing high-dimensional HSI data; otherwise, overfitting may occur.

To expand the number of training samples and enhance the representability of spectral–spatial features, researchers divide the image into multiple scales of superpixel images and directly use the boundaries of superpixels at different scales to precisely determine the category of each pixel while combining the multifeature approach to process several kinds of data features simultaneously. Kotzagiannidis and Schönlieb [35] constructed a graph structure and label propagation method for multifeature learning by embedding multiple superpixel features and multigraph learning to achieve optimal modeling with limited data. Yin and Cui [36] decomposed HSI into several Gaussian pyramids to extract multiscale (MS) features and constructed a Gaussian pyramid-based MS feature extraction method to effectively improve the classification accuracy while using limited samples. Duan *et al.* [37] constructed MS structural features of HSIs by the relative total variation method with different parameter settings to achieve accurate classification of small sample sizes with severe image noise. Sun *et al.* [38] proposed an adjacent superpixel-based MS spatial–spectral kernel method, which greatly improved the quality of training samples. All the above methods are based on multiple scales and features or their variants. To ensure that the most adequate spectral–spatial features are obtained with a small training sample, we propose a multi-structure network formed from MS and multifeature modules to comprehensively obtain the most representative features while being restricted by ultrasmall sample rates. To describe the direct differences between HSI pixels, we extract four types of features in the HSI: origin spectral features, extended multiple attribute profile (EMAP) features [22], local binary pattern (LBP) features [39], and 3-D-Gabor features [40]. We first extract the origin spectral features to exploit the information of spectral bands under different feature categories. Then, since our goal is to deal with the HSI classification task at ultrasmall sample rates, the diversity of features becomes particularly important. Thus, we extract information in terms of morphological attribute features, spatial texture features, and frequency-domain texture features. Specifically, EMAP features compute multiple morphological attributes in the image while reducing the dimensionality of the HSI image, LBP features effectively implement the description of spatial texture features by calculating the local gray-scale differences between the central pixel and its neighborhood pixels, and 3-D-Gabor features are able to find the best-combined localization in both frequency and spatial domains, considering texture information along with spectral features. They simultaneously exploit the spatial–spectral information of small samples from multiple perspectives. Since the MS and multifeature methods require running multiple classifiers at the same time, these methods put high demands on the running speed of the classifier. Although the deep learning network classifier has accurate classification results and high generalization ability, it needs to establish complex relationships between HSI pixels; hence, it is less computationally efficient. Therefore, in the current report, many classifiers that are capable of fast training are considered.

The extreme learning machine (ELM) [41] is a time-efficient classifier. The weights of hidden layer nodes in ELM are randomly generated or artificially set without updating. The learning process only calculates the output weight, so the ELM has a high learning rate and generalization ability. Compared with the deep learning network, it greatly improves the computational efficiency of classification. Liu *et al.* [42] developed an HSI classification method, TL-ELM, by taking advantage of the fast classification of the ELM, combining it with transfer learning, and preserving input weights and hidden biases. Cao *et al.* [43] constructed a multilayer sparse representation of the ELM, which improved the stability of the ELM with respect to parameter changes during propagation. However, the ELM still has three obvious shortcomings: sensitive to overfitting, insufficient global search capability, and relatively shallow representation of features. To address the first shortcoming, researchers used a kernel approach to improve the ELM. The kernel ELM (KELM) replaces the hidden layer of the ELM with a kernel function, effectively avoiding the overfitting problem that may occur in the learning process [41]. Second, Chen *et al.* [41] optimized the KELM with the gray wolf optimization algorithm to improve the global search ability and enhance the generalization ability of small samples. Finally, Li *et al.* [44] proposed a deep neural network-based KELM to improve the extraction of depth features and achieve the best generalization effect. The MS and multifeature method using a KELM have obvious advantages over other classifiers; therefore, we employ the KELM as the classifier. However, it still has difficulties in regards to multiple result fusion.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:54:29 UTC from IEEE Xplore. Restrictions apply.

In the field of MS or multifeature image processing, the fusion of multiple scales and multiple features has always been a complex task. For MS and multifeature HSI classification, the most common method is the decision fusion mechanism method, which selects the label with the greatest majority possibility by voting. In addition, probabilistic fusion based on the probabilistic output of the classifier, which conducts fusion by stacking individual feature vectors to generate new vectors for classification, is another more effective fusion method [45]. Zhang *et al.* [46] proposed an MS dense network (MSDN) that achieved deep feature extraction and reconfiguration of MS fusion for HSI classification. Xi *et al.* [47] proposed a weighted output layer fusion strategy (WOLFS), which creatively combined training and testing classification accuracy as fusion weights with decision fusion strategy, and achieved relatively good results. However, the simple selection of results based on the data is hardly a measure of the visual effect, and an attention mechanism can visually improve classification performance. The attention mechanism has been widely applied in the feature selection of HSI classification. For example, in [48], an MS attention mechanism (MS-AM) was developed to adaptively recalibrate the nonlinear interdependence among the features. Xu *et al.* [49] combined self-attention with mutual attention to fuse feature vectors of different samples to obtain more stable feature expressions. The application of self-attention mechanisms in feature selection is well established, but, for classification fusion, it does not consider the influence of each category's classification accuracy on the fusion result. Especially for the case of large data size and complex structure to be fused, it is difficult to fully exploit the characteristics of each fusion decision by the ordinary majority voting mechanism. For the MS segmentation task, the classification fusion is vulnerable to the influence of scale selection. Therefore, it is difficult to select the best classification features at different scales. To accurately resolve the fusion of the classification results for different features at different scales, we propose a weighted self-attention fusion strategy (WSAFS), which introduces the self-attention mechanism into classifier fusion while considering the classification accuracies of each category in a multibranch manner. The effectiveness of decision fusion is enhanced by using test accuracy as fusion weights. The self-attention mechanism is also utilized to fully consider the attribute characteristics of spatial features at different scales. WSAFS improves the accuracy and the visual representation in the classification map. Consequently, a fusion strategy with optimal visual effects and evaluation metrics is achieved.

In summary, the main contributions of the multi-structure KELM with attention fusion strategy (MSAF-KELM) algorithm proposed in this article are given as follows.

1) The MSAF-KELM proposes a multi-structure feature extraction module that can realize the extraction of multiple features at different scales, which promotes the exploration of MS and multifeature information in a single sample, effectively improving the classification accuracy of the KELM classifier in ultrasmall data samples.
2) The MSAF-KELM proposes a WSAFS, which combines the classification weights of each feature and scale under multiple structures with the self-attention mechanism to achieve an efficient and accurate fusion of multibranch KELM classification results.
3) The MSAF-KELM can deliver significant classification performance at particularly ultrasmall sample rates compared with both traditional and state-of-the-art methods. Evaluations of four famous hyperspectral datasets validate the efficiency of the algorithm.

The remainder of this article is organized as follows. Section II reviews the related work and elaborates on the proposed MSAF-KELM. Section III explains the experiments and analysis. Section IV summarizes this article and puts forward some opinions and suggestions for future research.

II. Materials and Methods

In this article, the main architecture of our proposed MSAF-KELM algorithm for HSI classification is shown in Fig. 1. It involves three main steps.

Step 1: Adaptively determining the number of superpixels by the inherent spatial size and spatial resolution of the dataset and generating superpixel images by entropy rate segmentation (ERS). The EMAP feature, LBP feature, and 3-D-Gabor feature extractions are performed on superpixel segmentation images at different scales.

Step 2: The origin spectral features, EMAP features, LBP features, and 3-D-Gabor features of M scales (totally $3M + 1$ features) are each put into $3M + 1$ independent KELM classifiers to separately produce the classification results.

Step 3: The output results of $3M + 1$ independent KELM networks are fused by the WSAFS to select the most representative results and generate the final classification map.

In the following, we will elaborate on these three parts of the architecture.

A. Superpixel Feature Extraction and Combination

1) Superpixel Image Generation at Multiple Scales: An HSI contains complex and diverse information, with different sizes and shapes of the surface object areas. It is difficult to obtain the most delicate and accurate spatial surface attributes by feature extraction at a single scale. Therefore, we segment the HSI into different scale superpixel images. The superpixel is composed of a series of small areas adjacent to each other, which have similar color, brightness, texture, or other characteristics [50]. These superpixels, as the basic unit of image processing algorithms, effectively eliminate some abnormal pixels. Because the spatial dimension and resolution of HSIs are usually different, the number of superpixels should also be different. Before calculating the exact number, we first calculate an approximate range S_r to limit the number of pixels in each superpixel

$$S_r = \text{floor}\left(100 \times \theta^{\sqrt{\text{Res}}}\right) \tag{1}$$

where floor is the operator that rounds down to the nearest integer, Res is the spatial resolution of an HSI, and θ is a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:54:29 UTC from IEEE Xplore. Restrictions apply.

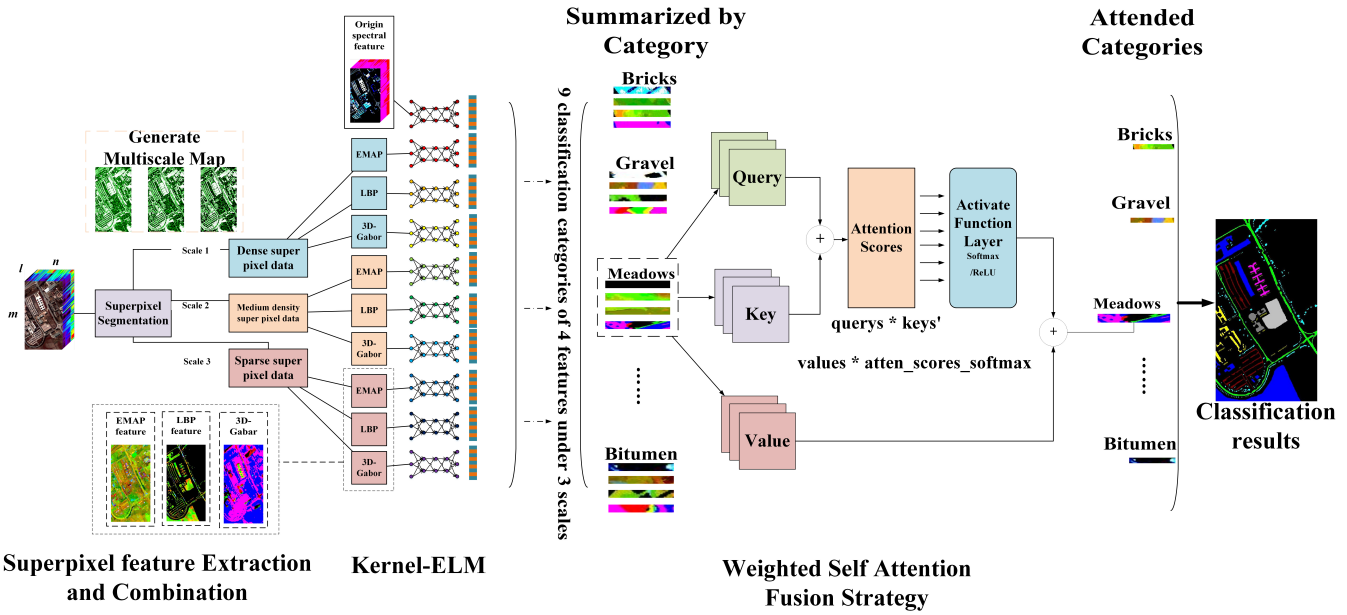


Fig. 1. Main architecture of our proposed MSAF-KELM algorithm.

parameter that controls monotonicity. According to the control range of the number of pixels in the superpixel, the number of superpixels at different scales can be calculated from the following formula [51]:

$$N_m = \text{floor}\left(\frac{R \times C}{S_r \times m}\right), \quad m = 1, \ldots, M \tag{2}$$

where m refers to the serial number of the scale, M refers to the maximum number of the scale, and R and C, respectively, refer to the height and width of the HSI, i.e., spatial size. After calculating the number of superpixels at different scales, we use the ERS algorithm to segment the image. Then, the image can be divided into uniform and compact superpixel regions, which is more in line with human eye perception, and each region has high similarity. We use the first PCA image of an HSI as the basic image of the segmented image. The ERS algorithm uses a weighted undirected graph to represent the image. Mathematically, given an undirected graph $G = (V, E)$, where V represents the vertex set of the graph and E represents the edge set of the graph, V and E represent pixels in the image and the similarity between pixels, respectively. The ERS algorithm divides V into a series of disjoint sets, in which the intersection of any two subsets is an empty set, and the union of all subsets is V. Then, we can select subset A of E and get an undirected graph $G' = (V, A)$. Therefore, the ERS algorithm can be transformed into the following formula:

$$\max H'(A) + \lambda T(A) \tag{3}$$

where $H'(A)$ is the random walk entropy rate in the graph, $T(A)$ represents the equilibrium term, A is the edge set, and λ is a nonnegative balance coefficient.

Fig. 2 shows the MS superpixel segmentation map of the Pavia University dataset. We average and normalize all pixels

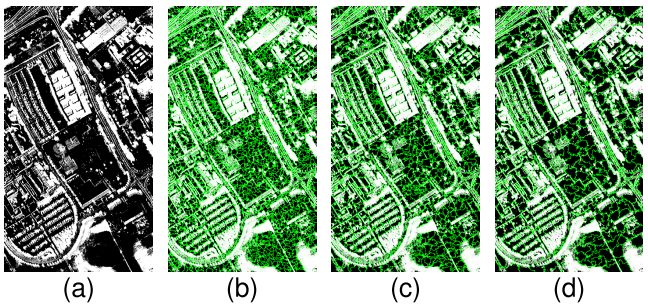


Fig. 2. First principal component map and MS superpixel segmentation map of the Pavia University dataset. (a) First PC map. (b) Dense superpixel map. (c) Medium density superpixel map. (d) Sparse superpixel map.

inside each superpixel. Given an HSI image $\mathbf{I} \in \mathbf{R}^{m*n*l}$, $\mathbf{I} = (p_1, p_2, \ldots, p_{m*n})$, where $m * n$ is the spatial resolution, l is the spectral dimension, and p_i is a pixel in the HSI image. Since each superpixel belongs to the same feature category, we uniformly take the average value of all pixels within each superpixel and treat it as one pixel. Given a superpixel sup_i, where the pixel set is $(e_1, e_2, \ldots, e_J) \in \text{sup}_i$, then each pixel e' is unified by

$$e' = \frac{\sum_{1}^{J} e_j}{J} \tag{4}$$

where J is the number of pixels in the superpixel sup_i.

2) EMAP Feature Extraction at Multiple Scales: After establishing MS superpixel segmentation images, we start to extract EMAP features, LBP features, and 3-D-Gabor features at each scale. EMAPs are higher level features extracted from high-resolution images. In the process of PCA, multiple attribute profiles (APs) are connected into one vector in the dimension reduction space. Thus, the accuracy and robustness

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:54:29 UTC from IEEE Xplore. Restrictions apply.

of spectral–spatial features are greatly improved. The AP decomposes the input image according to a set of ordered thresholds based on attribute filtering and obtains the AP of the image through a series of thickening and thinning operations. First, PCA is performed on the MS superpixel HSI to generate the image $\mathbf{I}_{\text{pca}}$, $\mathbf{I}_{\text{pca}} = (i_1^{m*n}, i_2^{m*n}, \ldots, i_b^{m*n})$, where b is the reduced spectral bands after PCA and i is the gray image represented by a component. The AP feature is extracted from each component i, and the different attributes in i, such as area, height, and volume, are integrated and connected to a superposition vector to obtain an accurate and delicate feature representation flexibly. Given an ordered threshold sequence $\{\lambda_1, \lambda_2, \ldots, \lambda_n\}$, an AP can be represented as follows:

$$\{\lambda_1, \lambda_2, \ldots, \lambda_n\}\text{AP}(i) = \{\phi_n(i), \ldots, \phi_1(i), i, \gamma_1(i), \ldots, \gamma_n(i)\} \quad (5)$$

where λ_i and γ_i, respectively, represent thickening and thinning transformations, and i represents a certain component. After generating the AP features of each component, we superimpose b components of the AP to generate EAP

$$\text{EAP} = \{\text{AP}(f_1), \text{AP}(f_2), \ldots, \text{AP}(f_b)\}. \quad (6)$$

We extract the EAP of area, inertia, and standard deviation at the same time, and cascade them into a vector. This method, which can reduce the dimension of the HSI while concatenating multiple attributes to extract its morphological AP features, is called extended morphological AP. At this point, the EMAP feature extraction at each scale is completed.

3) LBP Feature Extraction at Multiple Scales: Next, we continue to extract LBP features. The LBP mainly calculates the gray level of each central pixel and its domain pixels and represents the texture features of the image through their gray level difference. We first extract the first principal component i_1^{m*n} of an HSI as the basic 2-D image for extracting LBP features. Given a vertex $g_c \in i_1^{m*n}$ in the image as the central pixel, each pixel $\{g_i\}_{i=0}^{B-1}$ in its field is assigned a binary label, and B is the number of pixel members in the field after equidistant sampling of a circle with R as the radius. The value of radius R determines the spatial resolution of the domain. If the gray value of a pixel in the field is greater than the central pixel, the label of that pixel is 1; otherwise, it is 0. Therefore, the extraction of LBP features can be represented as follows:

$$\text{LBP}_B = \sum_{i=0}^{B-1} s_{\text{lbp}}(g_i - g_c)2^i \quad (7)$$

$$s_{\text{lbp}}(x) = \begin{cases} 1, & x \geqslant 0 \\ 0, & x < 0. \end{cases} \quad (8)$$

The LBP code of each central pixel in the HSI is calculated by the LBP operator. The histogram of LBP codes for different center pixels, i.e., the frequency of each number, is calculated. Finally, the statistical histogram of each region is connected to form a feature vector, which is the LBP feature vector of the whole HSI.

4) 3-D-Gabor Feature Extraction at Multiple Scales: To fully preserve the morphological texture features in the HSI and to show the signal variations in the spectral–spatial domain, we extract 3-D-Gabor features directly from the HSI data. The Gabor filter is a short-time Fourier transform with a Gaussian function as the window function. The 3-D-Gabor can find the best-combined localization by considering three frequency and spatial dimensions simultaneously to efficiently extract texture and spectral features in the hyperspectral cubic structure.

In the spectral–spatial frequency domain (x, y, z), the 3-D-Gabor data can be represented as follows:

$$\begin{aligned} G &= (x, y, z) \\ &= m(x, y, z) \times n(x, y) \times o(x, y, z, \varepsilon, \varphi) \\ m(x, y, z) &= \frac{1}{(2\pi)^{3/2}\sigma_x\sigma_y\sigma_z} \\ n(x, y) &= \exp\left(-\frac{1}{2}\left(\left(\frac{x'}{\sigma_x}\right)^2 + \left(\frac{y'}{\sigma_y}\right)^2 + \left(\frac{z'}{\sigma_z}\right)^2\right)\right) \\ o(x, y, z, \varepsilon, \varphi) &= \exp\left(i2\pi\left(x\varepsilon_x + y\varepsilon_y + b\varepsilon_z\right)\right) \end{aligned} \quad (9)$$

where $\varepsilon_x = \varepsilon\sin\varphi\cos\alpha$, $\varepsilon_y = \varepsilon\sin\varphi\cos\alpha$, and $\varepsilon_z = \varepsilon\cos\alpha$. σ denotes the width of the Gaussian envelope in each direction in the spectral–spatial frequency domain (x, y, z), which is used to determine the size of the Gaussian envelope. φ denotes the angle between the vector and the spectral dimension, and α is the angle between the projection of the wave vector in the spatial plane and the x-direction, which determines the direction of the center frequency of the 3-D-Gabor filter. (x, y, z) is the coordinate of a data point in the hyperspectral cube rotated by the φ and α angles. ε denotes the frequency, which determines the magnitude of the spatial scale factor. φ and α usually take values in the set $\{0, \pi/4, \pi/2, 3\pi/4\}$. When $\varphi = 0$, the three central frequency components are equal, so one of the combinations can represent all directions. Therefore, the total number of Gabor filters is 13. We input the raw hyperspectral cube data of the HSI with the size of (x, y, z) into a 3-D-Gabor filter for convolution, which constitutes a 3-D structured data with texture and spectral features

$$\text{HSI}_{\text{3D-Gabor}}(x, y, z) = \text{HSI}(x, y, z) \otimes G \quad (10)$$

where $\otimes$ denotes the convolution operator.

B. Kernel Extreme Learning Machine Autoencoder Module

Based on the above analysis, we can see that a large number of inputs will be generated after MS and multifeature processing of an HSI. Therefore, selecting an efficient and fast classifier is very important for the whole algorithm. If the classification efficiency of any scale or feature is not high, it will seriously affect the whole classification effect. KELM is a special feedforward neural network. It has only a simple single hidden layer node, which has great advantages in learning efficiency and generalization. Thus, we choose the KELM as our classifier network. Consider that the training feature sample set $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^N$ at a given scale m, where N is the number of features. Take the whole $\mathbf{X}$ set as the network input; then, the corresponding label set is $\mathbf{Y} = \{\mathbf{y}_i\}_{i=1}^L$, where L is the number of classification labels. First, let us look at the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:54:29 UTC from IEEE Xplore. Restrictions apply.

propagation mode of an ordinary ELM. Give the propagation mode of the jth layer of an ELM

$$f(\boldsymbol{x}_i) = \sum_{j=1}^{M} \boldsymbol{\gamma}_j h\left(< \boldsymbol{w}_j, \boldsymbol{x}_i > + b_j\right) = y_i \tag{11}$$

where $\boldsymbol{x_i}$ is the feature of a single input sample, M is the number of hidden nodes in this layer, $\boldsymbol{w_j}$ is the weight of the jth hidden node, and b_j is the bias of the jth hidden node. According to the characteristics of the ELM, they are randomly set values, and there is no need to adjust and update them after setting. Compared with the backpropagation of ordinary neural networks, the ELM greatly reduces the amount of calculation and improves classification efficiency. $< \cdot >$ is the point multiplication operation, h is the activation function, and $h(< \boldsymbol{w}_j, \boldsymbol{x}_i > + b_j)$ represents the whole hidden layer calculation process. $\boldsymbol{\gamma}_j$ is the weight vector connecting the jth hidden node and the output node, indicating the calculation process of the output classification of the hidden node. Formula (11) can also be expressed by the following formula:

$$\mathbf{H}\boldsymbol{\gamma} = \mathbf{Y} \tag{12}$$

where $\mathbf{Y}$ is the label matrix, $\boldsymbol{\gamma} = (\gamma_1, \gamma_2, \ldots, \gamma_M)$, and $\mathbf{H}$ is the implicit output node matrix. Therefore, it can also be transformed into a linear problem

$$\boldsymbol{\gamma} = \mathbf{H}^{\dagger}\mathbf{Y} \tag{13}$$

where $\mathbf{H}^{\dagger}$ is the Moore–Penrose generalized inverse of $\mathbf{H}$. $\mathbf{H}^{\dagger}$ can be calculated by $\mathbf{H}^{\dagger} = \mathbf{H}^{\mathrm{T}}(\mathbf{H}\mathbf{H}^{-1})^{-1}$. To improve the generalization ability, an ELM introduces the positive parameter ρ^{-1}, so the final propagation mode is

$$f(\boldsymbol{x}_i) = h(\boldsymbol{x}_i)\mathbf{H}^{T}\left(\frac{\mathbf{I}}{\rho} + \mathbf{H}\mathbf{H}^{\mathbf{T}}\right)^{-1}\mathbf{Y}. \tag{14}$$

Next, we use the kernel method to improve the ELM. Let $\Omega = \mathbf{H}\mathbf{H}^{\mathbf{T}}$, $\Omega = h(\boldsymbol{x}_i)h(\boldsymbol{x}_j) = K(\boldsymbol{x}_i, \boldsymbol{x}_j)$, where K is the kernel function satisfying Mercer's condition. Therefore, the output of the KELM can be expressed as

$$f_{\mathrm{KELM}}(\boldsymbol{x}_i) = \begin{bmatrix} \Omega_{i,1} \\ \Omega_{i,2} \\ \cdots \\ \Omega_{i,N} \end{bmatrix}^{T} \left(\frac{\mathbf{I}}{\rho} + \Omega\right)^{-1}\mathbf{Y}. \tag{15}$$

To more efficiently exploit the refined and detailed features, and deeply extract the important information contained in the features, we combine the autoencoder with the KELM network to carry out the classification task. At the same time, by stacking autoencoders, the network depth is extended to generate a deep KELM autoencoder network. As shown in Fig. 3, a total of N input matrices $\mathbf{X}$ generate c classification results through the implicit S layer kernel matrix Ω transformation. The matrix for transformation between each layer Ψ indicates that

$$\Omega\Psi = \mathbf{Y}. \tag{16}$$

The transformation matrix Ψ can be expressed by a propagation formula similar to (15)

$$\Psi = \left(\frac{\mathbf{I}}{\rho} + \Omega\right)^{-1}\mathbf{Y}. \tag{17}$$

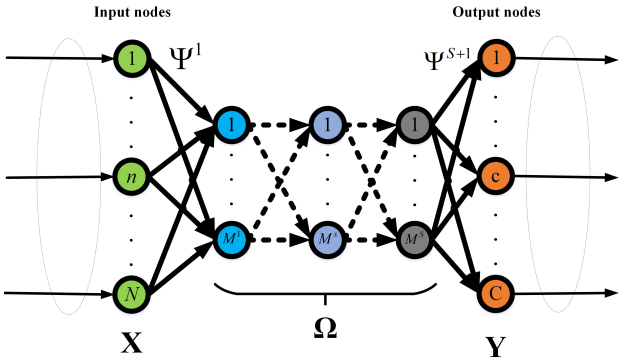


Fig. 3. Illustration of deep KELM-AE.

Finally, the propagation mode of the Sth layer deep KELM autocoder network can be expressed as

$$\mathbf{X}^{(s+1)} = h\left(\mathbf{X}^{(s)}\left(\Psi^{(s)}\right)^{\mathrm{T}}\right). \tag{18}$$

Different from the conventional deep learning neural network, deep KELM is a single forward training method without gradient descent backpropagation. Each layer of the transformation is obtained through the least square solution, which has the advantage of great training efficiency. We input the M scales EMAP features, LBP features, 3-D-Gabor features, and one origin spectral feature into $3M + 1$ deep KELM classifiers, and each classifier carries out a classification result separately.

C. Weighted Self-Attention Fusion Strategy

We have input the features at different scales into a total of $3M + 1$ deep KELM classifiers, so a total of $3M + 1$ classification result prediction maps are generated. Each prediction map has different characteristics at different scales and features, and each category also has different classification performance for different features due to their respective spatial morphology and spectral properties. It is a challenging task to grasp the focus at each scale and feature, and fuse these classification prediction maps into one map while maintaining maximum classification accuracy. To solve this problem, we propose a WSAFS. We combine the output labels of training samples with the self-attention mechanism to simulate the attention mechanism of human vision to selectively focus on the more interesting classification label information while utilizing limited visual processing resources. WSAFS can be expressed as

$$A(p, n) = \max_{n=[1,2,\ldots,N]} C(n) \tag{19}$$

$$C(n) = \sum_{k=1}^{3M+1} \mathbf{TR}_n^K \cdot \mathbf{TE}_n^K \cdot \mathbf{SA}^K \tag{20}$$

where $A(n)$ refers to the final prediction label for the input pixel p. $C(n)$ refers to the number of classifiers whose final prediction result belongs to the nth category. It is determined by three parts: the training accuracy, the test accuracy, and the weight of self-attention of each classifier. $\mathbf{TR}_n^K$ refers to the accuracy of the kth classifier for the nth category on the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:54:29 UTC from IEEE Xplore. Restrictions apply.

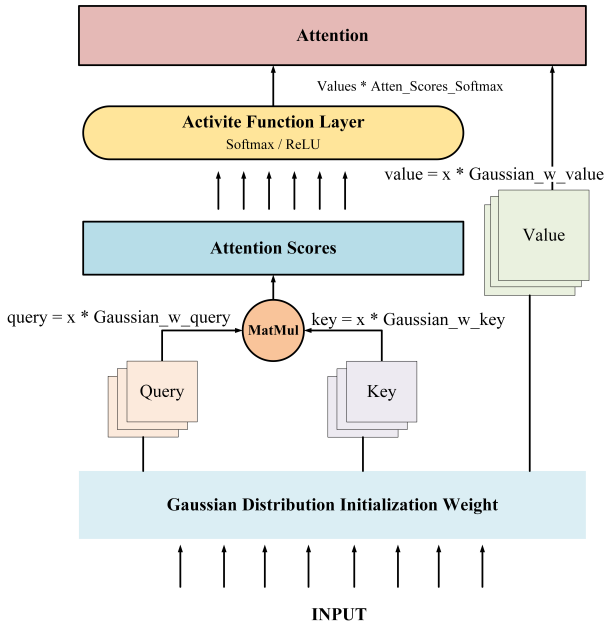


Fig. 4. Illustration of self-attention.

training samples, and practically, the value of $\mathbf{TR}_n^K$ always tends to 1. $\mathbf{TE}_n^K$ refers to the accuracy of the kth classifier for the nth category of the test samples. $\mathbf{SA}^K$ refers to the adaptive weight of the self-attention operation on the classification result of the kth classifier. It interacts the output information of the classifiers with each other and finds the objects they should pay more attention to. To be more specific, the structure of self-attention is depicted in Fig. 4. The classification weight of self-attention is specifically expressed as follows:

$$\mathbf{SA} = \text{softmax}(\mathbf{Attention})\ \mathbf{V} \tag{21}$$

$$\mathbf{Attention} = \mathbf{Attention_Scores}(\mathbf{Q}, \mathbf{K}). \tag{22}$$

Among them, $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ represent Query, Key, and Value, respectively. They are the data generated by three types of learnable weight matrices. They constantly modify and update the input information to obtain the attention relevance of the input vector. Given the dataset $\mathbf{D} = \{d_i\}_{i=1}^{3M+1}$ to input into the self-attention mechanism, then the three types of learnable weight matrices initialized with Gaussian distributions are multiplied by the input data as follows:

$$\begin{aligned} \mathbf{K} &= \mathbf{W^K D} \\ \mathbf{Q} &= \mathbf{W^Q D} \\ \mathbf{V} &= \mathbf{W^V D} \end{aligned} \tag{23}$$

where $\mathbf{W^K}$, $\mathbf{W^Q}$, and $\mathbf{W^V}$ are the weights initialized with a random Gaussian distribution. We use a 2-D Gaussian distribution to initialize the weight matrix and specify a Gaussian distribution with a mean of 0 and a standard deviation of 0.5 to guide the feature distribution to be more regular. Then, we use the obtained Q and K for point multiplication to calculate the correlation between each two input vectors and essentially to calculate the attention scores matrix

$$\mathbf{Attention_Scores} = \mathbf{K}^{\mathrm{T}} \times \mathbf{Q} \tag{24}$$

where T is the transpose symbol. Each value in the matrix **Attention_Scores** records the strength of the attention corresponding to the two input vectors. Then, we perform the softmax operation or the ReLU operation on the attention matrix to obtain $\mathbf{A}$. After using the obtained $\mathbf{A}$ and $\mathbf{V}$ to calculate the weight of each input vector s corresponding to self-attention and the output vector SA, we have

$$\mathbf{SA} = \mathbf{V} \times \mathbf{A}. \tag{25}$$

To sum up, WSAFS not only considers the classification differences of each category with different scales and features but also cleverly combines the self-attention mechanism to accurately fuse and summarize the prediction maps under each classifier. The overall process of the proposed MSAF-KELM method is shown in Algorithm 1.

Algorithm 1 MSAF-KELM Model

Require:
HSI data $\mathbf{I} \in \mathbf{R}^{m*n*l}$, ground-truth $\mathbf{Y} \in \mathbf{R}^{m*n}$; scale number M; training sample rate $\mu\%$.

Ensure:
Predicted labels of the test dataset.

1: Determine the number of superpixels of M scales based on the size and resolution of the $\mathbf{I}$.
2: ERS operation for the first PC image.
3: Average uniformization of all pixel values within each superpixel.
4: Obtain superpixel EMAP features on the first three PCs.
5: Obtain superpixel LBP features on the first PC.
6: Obtain superpixel 3D-Gabor features on the first 30 PCs.
7: **for** $i = 1$ to $3M + 1$ **do**
8: Perform deep KELM-AE classification operation and get the classification training weights **TR** and testing weights **TE**.
9: **end for**
10: Initialize the classifier output to $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ with Gaussian distribution.
11: $\mathbf{Q}$ and $\mathbf{K}$ are multiplied to obtain the attention scores.
12: Operate $softmax$ function on attention scores.
13: Multiply the softmax result with $\mathbf{K}$ to get the attention weight **SA**.
14: Use **TR**, **TE**, and **SA** to jointly decide the final prediction label and achieve fusion results.

III. Experiment and Analysis

In order to verify the effectiveness of our proposed MSAF-KELM method for HSI classification, we selected four kinds of hyperspectral datasets, including the Salinas, Pavia University, Botswana, and Xuzhou datasets. In addition, we choose several classic and state-of-the-art hyperspectral classification methods to compare with our proposed method, including RF [52], multivariate linear regression (MLR) [53], SVM [20], multiple layer perceptron (MLP) [54], CNNs [55], HybridSN [29], MS context-aware ensemble deep KELM (MSC-EDKELM) [47], dual-weighted KELM (DW-KELM) [56], and spectral–spatial feature tokenization transformer

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:54:29 UTC from IEEE Xplore. Restrictions apply.

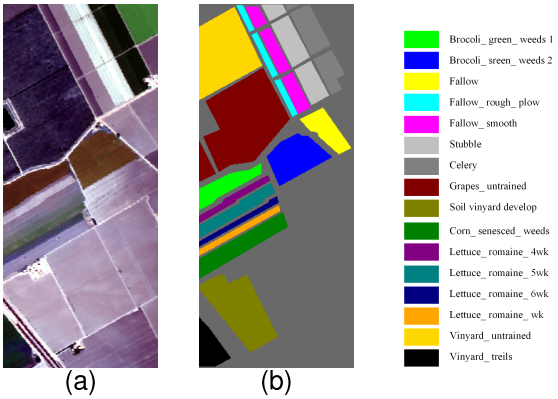


Fig. 5. Salinas dataset. (a) False-color map. (b) Ground-truth map.

TABLE I
TRAINING AND TEST SAMPLES IN THE SALINAS DATASET

ID	Land Cover Class	Training	Test
C01	Alfalfa	11	1998
C02	Corn-notill	19	3707
C03	Corn-mintill	10	1966
C04	Corn	7	1387
C05	Grass-pasture	14	2664
C06	Grass-trees	20	3939
C07	Grass-pasture-mowed	18	3561
C08	Hay-windrowed	57	11214
C09	Oats	32	6171
C10	Soybean-notill	17	3261
C11	Soybean-mintill	6	1062
C12	Soybean-clean	10	1917
C13	Wheat	5	911
C14	Woods	6	1064
C15	Vinyard_untrained	37	7231
C16	Vinyard_vertical_trellis	10	1797
	Total	279	53850

(SSFTT) [57]. To demonstrate the efficient classification performance of the MSAF-KELM for ultrasmall sample rates, we set the training sample rate of the Salinas, Pavia University, and Xuzhou datasets to 0.5% and Botswana to 5% in the comparison experiments.

A. Data Description

1) Salinas: First, the Salinas dataset is an HSI captured by an Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensor in Salinas Canyon, California. The image consists of 512 × 217 pixels, and its spatial resolution is 3.7 m. The original data contain 224 spectral bands, but, in our experiment, we discarded 20 water vapor absorption bands, leaving 204 bands. The ground truth of this dataset contains 16 crop categories. The specific labels and the division of the training and test sets are shown in Fig. 5 and Table I.

2) Pavia University: The second dataset is the scene taken by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor in Pavia, Northern Italy. The image consists of 610 × 340 pixels, and its spatial resolution is 1.3 m. The original data contain 115 spectral bands. After removing some invalid bands, we choose 103 bands for our experiment. The ground truth of this dataset contains nine categories. The

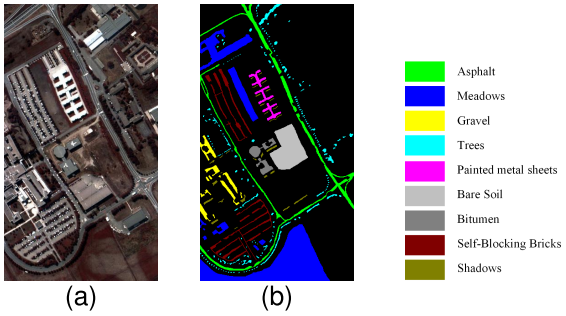


Fig. 6. Pavia University dataset. (a) False-color map. (b) Ground-truth map.

TABLE II
TRAINING AND TEST SAMPLES IN THE PAVIA UNIVERSITY DATASET

ID	Land Cover Class	Training	Test
C01	Asphalt	34	6567
C02	Meadows	94	18525
C03	Gravel	11	2058
C04	Trees	16	3018
C05	Painted metal sheets	7	1308
C06	Bare Soil	25	4974
C07	Bitumen	7	1293
C08	Self-Blocking Bricks	19	3633
C09	Shadows	5	912
	Total	213	42293

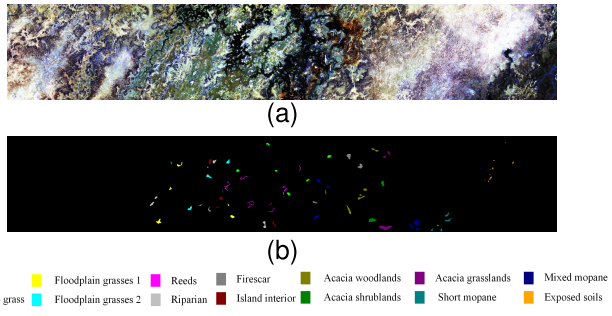


Fig. 7. Botswana dataset. (a) False-color map. (b) Ground-truth map.

specific labels and the division of the training and test sets are shown in Fig. 6 and Table II.

3) Botswana: The third dataset is the hyperspectral dataset taken by the Hyperion sensor on the EO-1 satellite from the National Aeronautics and Space Administration (NASA) over Okavango, Botswana, Africa. The image consists of 1476 × 256 pixels and covers 242 bands of 400–2500 nm. Some water vapor absorption bands and other noise bands are removed, and 145 bands are selected for the experiment. The ground truth of this dataset contains 14 categories. The specific labels and the division of the training and test sets are shown in Fig. 7 and Table III.

4) Xuzhou: The fourth dataset is the HSI collected from an urban and rural location in Xuzhou, Jiangsu, People's Republic of China, in November 2014. The image is composed of 500 × 260 pixels and has a high spatial resolution of 0.73 m. After removing the noise band of 415–2508 nm, there are 436 spectral bands left. High spatial resolution and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:54:29 UTC from IEEE Xplore. Restrictions apply.

TABLE III
TRAINING AND TEST SAMPLES IN THE BOTSWANA DATASET

ID	Land Cover Class	Training	Test
C01	Water	14	256
C02	Hippo grass	6	95
C03	Floodplain grasses 1	13	238
C04	Floodplain grasses 2	11	204
C05	Reeds	14	255
C06	Riparian	14	255
C07	Firescar	13	246
C08	Island interior	11	192
C09	Acacia woodlands	16	298
C10	Acacia shrublands	13	235
C11	Acacia grasslands	16	289
C12	Short mopane	10	171
C13	Mixed mopane	14	254
C14	Exposed soils	5	90
	Total	170	3078

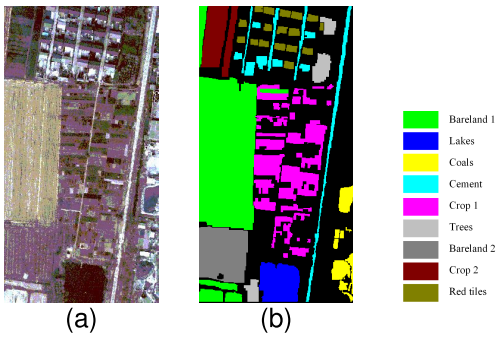


Fig. 8. Xuzhou dataset. (a) False-color map. (b) Ground-truth map.

TABLE IV
TRAINING AND TEST SAMPLES IN THE XUZHOU DATASET

ID	Land Cover Class	Training	Test
C01	Bareland 1	132	26264
C02	Lakes	21	4006
C03	Coals	14	2769
C04	Cement	27	5187
C05	Crops 1	66	13118
C06	Trees	13	2423
C07	Bareland 2	35	6955
C08	Crop 2	24	4753
C09	Red tiles	16	3054
	Total	348	68529

complex land-cover distributions make it difficult to classify the Xuzhou dataset. The ground truth of this dataset contains nine categories. The specific labels and the division of the training and test sets are shown in Fig. 8 and Table IV.

B. Experimental Setting

1) Evaluation Criteria: To accurately analyze the performance of our proposed method and compare it with other methods, we use four evaluation indicators: overall accuracy (OA), average accuracy (AA), kappa coefficient (K), and classification accuracy of each category. The higher the value of the four evaluation indicators, the better the classification performance of the method.

2) Environment Configuration: The method that we proposed is carried out in the MATLAB R2018b environment. For the methods used for comparison, we run all experiments in the PyTorch environment. We use an Intel Xeon silver 4210 CPU, 128-GB RAM, and an NVIDIA Geforce RTX 2080ti 11-GB GPU server. For the comparison methods of deep learning, the original learning rate is set to $1e-3$, and 100 training epochs are used in each experimental training model.

3) Parameter Setting Adjustment: In the parameter analysis, we analyze several parameters that affect the classification performance and training process, including the scale number of superpixel segmentation, the combination form between different features in feature extraction, and the number of kernel layers in the deep KELM. We chose 30 labels of each category from the Salinas, Pavia University, and Xuzhou datasets and 15 labels from each category on the Botswana dataset as the training sets for the experiments.

Fig. 9 shows the impact of different scales on the classification results for the Salinas, Pavia University, Botswana, and Xuzhou datasets. In Fig. 9, the x-axis represents the number of scales for superpixel segmentation, and the y-axis represents the OA of classification. As seen from the four subfigures, too many or too few scales cause lower classification performance. Too few scales lead to insufficient accuracy of the extracted features. Too many scales lead to insufficient generalization ability and overfitting phenomenon, which seriously affects the classification accuracies finally. When the number of scales is 3 for all four datasets, our proposed method (i.e., MSAF-KELM) achieves the optimal classification performance.

Fig. 10 shows the impact of different combinations of HSI features on the classification results for the Salinas, Pavia University, Botswana, and Xuzhou datasets. From these four subfigures, we can see that different combinations of features lead to wildly different classification results because each feature contains different spectral or spatial feature attributes. The combination of Origin Spectral + LBP, Origin Spectral + EMAP, and Origin Spectral + 3-D-Gabor in the four subfigures shows that the morphological features and spatial domain texture frequency features are more sensitive to the spatial features. However, after removing the LBP features, the Origin Spectral + EMAP + 3-D-Gabor combination does not achieve the highest classification accuracy. Second, if the combination of LBP + EMAP + 3-D-Gabor with the origin spectral features removed is chosen, it does not work as well on the Botswana dataset. Therefore, for datasets with more complex spectral dimensions, the original spectral features can significantly improve the classification accuracy. After summarizing four different datasets and considering the classification effects and computational efficiency of the corresponding features, we decide to use the combination of origin spectral features, LBP features, EMAP features, and 3-D-Gabor features so that our proposed method can achieve optimal classification performance.

Fig. 11 shows the impact of the number of network layers of the deep KELM classifier for the Salinas, Pavia University, Botswana, and Xuzhou datasets on the classification results. In Fig. 11, the x-axis represents the number of network layers

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:54:29 UTC from IEEE Xplore. Restrictions apply.

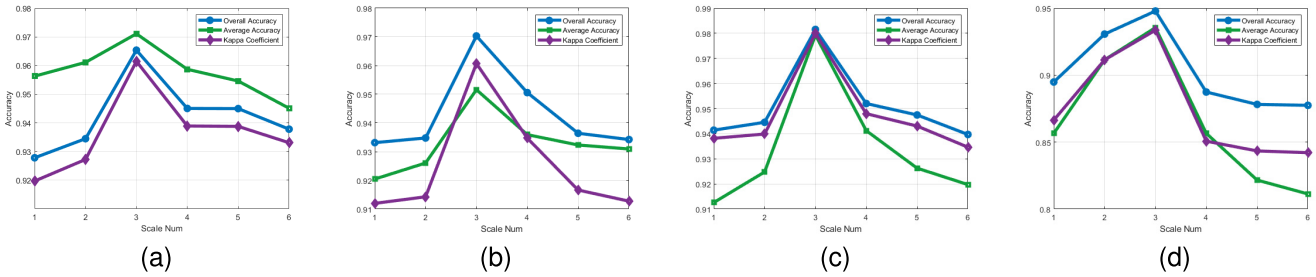


Fig. 9. Influence of different scale numbers on classification accuracy. (a) Salinas dataset. (b) Pavia University dataset. (c) Botswana dataset. (d) Xuzhou dataset.

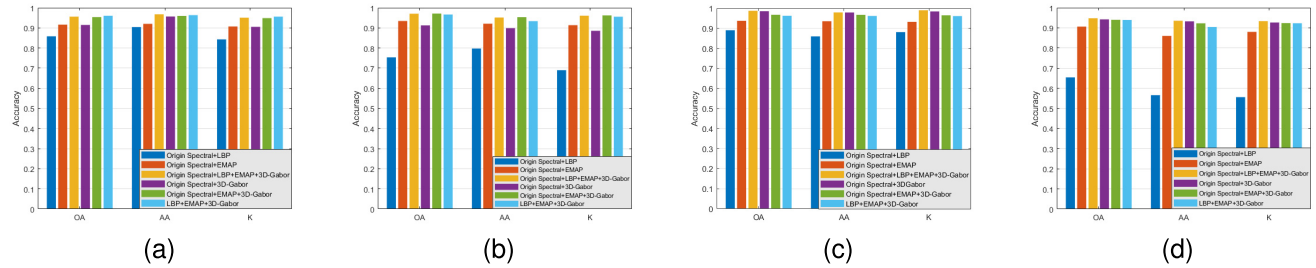


Fig. 10. Influence of different HSI feature types on classification results. The feature combination of data from (left) to (right) is: Origin Spectral + LBP, Origin Spectral + EMAP, Origin Spectral + LBP + EMAP + Gabor, Origin Spectral + Gabor, Origin Spectral + EMAP + Gabor, and LBP + EMAP + Gabor. (a) Salinas dataset. (b) Pavia University dataset. (c) Botswana dataset. (d) Xuzhou dataset.

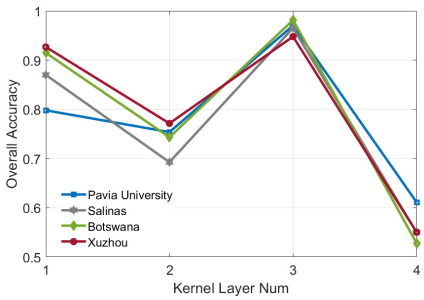


Fig. 11. Influence of the number of network layers of the KELM on classification accuracy.

TABLE V
ABLATION EXPERIMENTAL RESULTS

Cases	Component				Indicators		
	MS	MfC	KELM	WSAFS	OA(%)	AA(%)	$k * 100$
1	×	✓	✓	✓	92.78	95.63	91.97
2	✓	×	✓	✓	91.63	92.03	90.71
3	✓	✓	×	✓	86.99	92.04	85.51
4	✓	✓	✓	×	94.81	95.73	94.23
5	✓	✓	✓	✓	**96.54**	**97.11**	**96.15**

of the KELM classifier, and the y-axis represents the OA of classification. From Fig. 11, we can see that, if the network layers are too shallow, the network is grossly underutilized for features, and if the network layers are too deep, there will be severe data overfitting phenomenon. Whether the number of network layers is too high or too low, it is not conducive to the efficiency of classification. On average, when the number of network layers is 3, our proposed method achieves the maximum classification performance.

C. Ablation Experiment

To effectively demonstrate the classification performance of our method, we perform ablation experiments on the different modules of the method using the Salinas dataset. 30 labels from each category in the dataset are used for training, and the remaining labels are used for testing. We divide the proposed MSAF-KELM method into four modules, and use OA, AA, and K indicators to evaluate the impact of different modules on the final classification accuracy. Those modules are the MS module, the multifeature combination (MfC) module, the KELM autocoding module, and the WSAFS module. All experimental results are shown in Table V.

Case_1 cancels the MS module, and all experiments are carried out at a single scale of 3. At this time, the classification accuracies of OA, AA, and K are 3.76%, 1.48%, and 4.18% lower than those of the proposed MSAF-KELM method, respectively.

Case_2 cancels the MfC module. We only use the EMAP feature and origin spectral feature. At this time, the classification accuracies are 4.91%, 5.08%, and 5.44% lower in terms of OA, AA, and K indicators than those of the proposed MSAF-KELM method.

Case_3 is an autocoding module that removes the KELM. We directly use a one-layer network for the classification task. The number of network layers is shallow, and its generalization ability is worse. Specifically, the classification accuracies decreased the most compared to the original experiment, such that the OA, AA, and K decrease by 9.55%, 5.07%, and 10.64%, respectively.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:54:29 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
CLASSIFICATION PERFORMANCE OF THE SALINAS DATASET OBTAINED BY DIFFERENT METHODS (THE BEST RESULTS ARE SHOWN IN BOLD)

No.	Random Forest [52]	MLR [53]	SVM [20]	MLP [54]	CNN [55]	HybridSN [29]	DW-KELM [56]	MSC-EDKELM [47]	SSFTT [57]	MSAF-KELM
1	96.87±1.42	97.01±1.06	96.09±2.49	95.26±3.82	87.59±11.04	98.55±1.57	99.20±0.11	**100.00±0.00**	99.92±0.16	99.36±0.59
2	99.83±1.02	99.34±0.82	94.88±1.21	98.04±1.00	99.82±0.11	93.27±9.30	99.19±0.13	99.49±0.03	**99.99±0.02**	99.24±0.25
3	84.14±6.54	87.63±8.87	45.88±11.45	89.04±7.11	90.83±5.21	96.34±1.15	99.13±0.24	99.90±0.04	**99.99±0.02**	99.43±0.28
4	93.27±4.58	98.09±0.78	96.94±0.98	**98.44±0.50**	97.53±1.13	89.83±2.54	94.34±1.17	88.97±4.76	96.45±2.98	90.04±3.15
5	94.66±2.31	95.56±4.45	97.42±1.01	95.10±2.02	97.45±1.29	93.61±7.29	95.83±2.41	80.34±5.23	**98.86±0.36**	96.22±4.08
6	97.74±1.84	99.62±0.30	98.57±0.58	99.63±0.12	99.31±0.33	99.37±0.83	98.45±0.41	98.10±1.31	**99.86±0.09**	99.11±0.58
7	97.66±1.01	99.36±0.19	99.30±0.10	99.39±0.10	99.40±0.25	98.39±1.49	**99.78±0.10**	99.75±0.12	98.94±0.5	99.63±0.34
8	82.99±3.70	82.61±3.33	94.23±5.14	82.86±1.90	79.50±14.56	94.20±0.76	81.23±3.12	96.75±2.19	92.64±1.63	**97.51±0.58**
9	98.60±0.69	99.20±0.28	99.02±0.12	98.64±0.93	99.18±0.63	99.86±0.12	99.76±0.21	99.90±0.02	99.98±0.02	**100.00±0.00**
10	74.51±5.78	82.15±5.42	72.05±5.66	85.13±2.93	86.05±5.80	95.60±3.96	93.96±2.11	84.24±6.35	**97.99±0.98**	94.75±3.64
11	75.88±7.09	82.94±7.20	0.24±0.73	79.61±12.58	88.72±5.37	94.92±3.46	98.48±0.53	99.81±0.01	**99.98±0.04**	88.94±2.58
12	94.71±5.69	**99.29±0.44**	96.93±1.34	99.00±0.74	98.88±1.53	86.70±4.89	95.76±2.91	88.84±8.21	97.61±3.09	97.66±3.12
13	94.99±5.08	95.75±3.71	98.62±0.35	95.28±3.76	98.85±0.60	91.07±1.45	**99.67±0.25**	97.59±2.03	94.4±5.62	90.11±9.66
14	89.78±3.29	87.54±2.27	87.82±2.52	89.03±2.46	87.74±3.53	83.69±18.62	93.36±3.18	60.38±10.86	**95.17±2.71**	89.77±6.16
15	44.88±7.63	50.36±4.17	8.87±12.58	48.48±7.46	58.41±16.20	89.24±5.93	90.31±2.86	92.45±3.54	89.92±1.32	**93.02±4.20**
16	85.42±8.39	89.42±4.06	66.11±4.25	89.36±5.80	82.7±10.48	96.98±2.62	99.33±0.58	91.49±5.16	**99.48±0.29**	95.38±3.65
OA(%)	84.45±0.88	86.56±1.03	78.45±0.85	86.31±0.94	87.02±1.74	94.54±3.01	93.45±1.23	94.38±0.54	96.47±0.56	**96.77±0.09**
AA(%)	87.87±1.16	90.37±1.35	78.31±1.03	90.14±0.89	90.75±1.40	93.85±3.78	96.11±0.47	92.37±1.23	**97.57±0.35**	95.64±0.52
k×100	82.61±0.99	84.99±1.15	75.69±1.02	84.70±1.06	85.51±1.92	93.92±3.36	92.72±1.91	93.74±0.12	96.07±0.62	**96.40±0.11**

Case_4 removes the WSAFS module. We directly used the average accuracies of the multibranch classifiers during training and testing as the fusion results. At this point, although the classification is better than in the previous cases, it still decreases in terms of OA, AA, and K by 1.73%, 1.38%, and 1.92% compared with the proposed MSAF-KELM method. Therefore, the experimental results verify that our model has an improved classification effect.

D. Classification Results

In order to validate the effectiveness of our method for efficient classification with ultrasmall sample rates, we select some representative classification methods for comparison, including RF [52], MLR [53], SVM [20], MLP [54], CNN [55], HybridSN [29], DW-KELM [56], MSC-EDKELM [47], and SSFTT [57].

The first three methods above are classic classification methods based on machine learning. Then, we compare four deep learning methods, i.e., MLP, CNN, HybridSN, and SSFTT, and two similar classification algorithms, i.e., DW-KELM and MSC-EDKELM. The first is the MLP, which is a feedforward artificial neural network with low training complexity. Then, the CNN includes five layers, including the input layer, the convolution layer, the pooling layer, the full connection layer, and the output layer. HybridSN, DW-KELM, MSC-EDKELM, and SSFTT are four state-of-the-art HSI classifiers, where the network settings were set as described in their corresponding references.

Tables VI–IX list the experimental classification results of our proposed method and comparison methods using four datasets. To ensure the validity and generalizability of the experimental results, we repeated each experiment ten times and then took the average value and standard deviation. Finally, for the convenience of comparison, we bold the optimal values of accuracies for each class.

For the Salinas dataset, as shown in Table VI, we took an ultrasmall sample rate of 0.5% as the training set. Whether from the individual classification accuracy of each category or the OA and K, our classification performance was the best. Compared with the cutting-edge HSI classification method HybridSN, our method was 2.23%, 3.27%, and 2.66% higher in OA, AA, and K, respectively. Compared with DW-KELM, our method was superior to it under all evaluation criteria. Compared with the similar MS method, MSC-EDKELM, the performance of our method was not inferior. Our method obtained richer feature information and a stronger ability to extract spatial information of complex features. In addition, even though the MSAF-KELM has more features than MSC-EDKELM, which leads to greater difficulty in feature fusion, our method still achieved 2.39% higher OA. This demonstrates the excellent performance of our feature and scale fusion strategy. However, since we set a very small rate of training samples, it would lead to fewer samples available for training in some categories with small sample totals, which led to severe sample imbalance. SSFTT had a better ability to handle the unbalanced sample distribution, especially for the ultrasmall sample rates. Therefore, the classification accuracies for categories 13 and 14 (i.e., wheat and woods) were higher than those of the proposed method, which affected the value of the AA. On the other hand, the OA and kappa coefficient considered the overall number of samples, so they had less effect on OA and K.

Table VII shows the quantitative experimental results on the Pavia University dataset. For this dataset, we also took the ultrasmall sample rate of 0.5% as the training set, and the remaining sample data were used for the test. For the comprehensive evaluation indexes, OA and K, the performance of our proposed method was the best. Because the number of training samples was small and the complexity of the dataset was relatively gentle, the advantages of the deep learning methods were not too great compared with traditional machine learning methods. Specifically, the HybridSN method achieved better performance. Compared with DW-KELM, our method had better performance at small sample rates. The MSC-EDKELM method delivered an efficient performance for ultrasmall sample classification, which ranked second best of the comparison methods based on OA and K. Due to the sufficient hyperspectral feature information and the more accurate WSAFS fusion mechanism, our proposed method had an absolute advantage for the accuracy of OA and K.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:54:29 UTC from IEEE Xplore. Restrictions apply.

TABLE VII

CLASSIFICATION PERFORMANCE OF THE PAVIA UNIVERSITY DATASET OBTAINED BY DIFFERENT METHODS (THE BEST RESULTS ARE SHOWN IN BOLD)

No.	Random Forest [52]	MLR [53]	SVM [20]	MLP [54]	CNN [55]	HybridSN [29]	DW-KELM [56]	MSC-EDKELM [47]	SSFTT [57]	MSAF-KELM
1	82.03±7.69	85.25±3.22	82.41±3.77	83.23±3.72	90.52±5.63	88.76±3.21	**95.47±2.41**	89.30±2.21	92.67±3.28	92.81±6.86
2	91.69±2.78	93.78±1.90	93.71±1.84	95.99±1.59	98.39±0.79	98.98±0.62	99.50±1.13	99.36±1.84	98.9±1.3	**99.63±0.15**
3	31.88±13.94	55.29±8.96	59.95±9.25	29.61±17.91	57.74±16.24	68.89±3.03	**92.15±2.54**	84.01±5.21	79.75±4.15	85.34±7.85
4	75.15±8.28	78.14±5.21	82.17±3.02	75.20±3.31	70.02±8.22	74.51±8.74	82.29±1.59	**91.44±4.94**	83.9±3.69	87.61±7.88
5	98.19±2.36	95.16±7.79	94.78±10.89	96.15±4.18	98.64±1.06	92.76±6.51	97.46±0.37	**99.93±0.23**	99.97±0.06	96.39±4.45
6	38.46±7.03	61.40±3.62	70.73±4.23	36.51±10.82	37.78±7.46	73.80±8.56	98.90±1.65	**99.94±0.02**	96.89±1.52	99.12±0.74
7	53.08±15.23	29.94±10.41	73.51±13.34	35.41±17.38	22.16±28.61	94.65±2.43	77.78±2.33	83.14±5.64	**99.94±0.09**	80.99±6.05
8	75.96±7.25	80.53±4.35	78.24±4.68	82.47±5.96	70.85±13.49	78.77±5.81	89.79±3.21	**96.26±2.83**	80.4±9.47	93.57±5.82
9	99.32±0.27	98.94±0.80	**99.62±0.19**	98.43±1.05	99.58±0.56	62.03±19.13	41.08±19.41	27.07±21.19	86.16±10.17	49.43±10.53
OA(%)	77.63±0.98	82.68±1.34	84.98±1.32	79.29±1.38	81.31±1.42	88.32±2.21	94.34±1.42	94.20±1.21	93.88±1.21	**94.64±0.16**
AA(%)	71.75±2.06	75.38±1.41	81.68±2.23	70.33±3.28	71.74±3.99	81.46±4.59	86.05±1.32	85.61±0.41	**90.95±2.15**	87.21±3.18
k×100	69.70±1.28	76.72±1.70	79.93±1.76	71.58±2.04	74.06±2.12	84.24±3.08	92.45±0.32	92.28±0.64	91.87±1.61	**92.85±0.28**

TABLE VIII

CLASSIFICATION PERFORMANCE OF THE BOTSWANA DATASET OBTAINED BY DIFFERENT METHODS (THE BEST RESULTS ARE SHOWN IN BOLD)

No.	Random Forest [52]	MLR [53]	SVM [20]	MLP [54]	CNN [55]	HybridSN [29]	DW-KELM [56]	MSC-EDKELM [47]	SSFTT [57]	MSAF-KELM
1	98.75±1.55	99.96±0.12	99.34±0.96	**100.00±0.00**	**100.00±0.00**	99.22±0.21	89.06±2.45	96.09±2.11	**100.00±0.00**	98.24±2.26
2	76.25±9.49	93.02±4.73	59.06±20.19	86.25±9.03	82.41±11.68	69.94±9.22	52.08±9.12	98.96±1.23	**100.00±0.00**	96.35±2.22
3	88.32±4.05	98.57±1.21	93.99±2.95	96.89±1.70	99.35±1.01	99.50±2.12	**100.00±0.00**	**100.00±0.00**	87.93±10.54	99.83±0.29
4	84.31±5.91	95.44±1.63	72.94±23.97	86.86±4.04	91.83±2.83	82.19±3.15	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	98.68±4.93
5	72.34±7.15	81.87±2.93	77.43±5.97	81.80±5.01	84.67±6.01	98.02±1.32	87.84±2.19	91.37±3.67	**98.27±0.47**	94.86±3.78
6	52.24±5.72	68.91±5.27	48.39±15.80	67.07±8.68	75.93±10.86	94.53±4.56	96.47±2.65	**99.61±0.13**	97.73±1.17	97.25±2.98
7	91.06±4.63	94.35±4.73	93.29±2.60	97.52±0.86	97.74±0.79	99.60±0.25	99.59±0.14	98.37±1.21	**100±0.00**	**100.00±0.00**
8	86.11±6.74	96.37±2.51	90.16±4.63	95.39±2.88	93.21±7.78	92.72±2.76	84.97±3.49	**100.00±0.00**	**100±0.00**	98.91±3.32
9	68.93±6.95	87.08±4.73	75.37±7.09	83.26±4.74	75.39±15.86	99.00±0.34	96.64±2.45	98.32±0.19	**100±0.00**	99.13±1.82
10	69.15±10.02	85.55±5.71	73.64±3.70	70.38±5.45	85.78±9.02	97.50±2.31	**100.00±0.00**	**100.00±0.00**	99.83±0.34	99.92±0.08
11	91.45±6.19	92.83±3.99	88.52±5.05	94.55±1.98	90.96±4.53	98.95±0.11	96.90±1.64	97.59±1.03	94.74±4.6	**99.55±0.75**
12	85.58±3.76	92.15±2.65	94.48±2.27	94.53±3.18	88.37±9.07	98.29±1.56	**100.00±0.00**	88.95±8.23	69.88±1.66	**98.78±2.90**
13	78.35±5.26	88.71±3.29	77.49±2.78	76.16±7.27	91.55±5.25	85.28±7.12	**100.00±0.00**	99.61±0.02	**100±0.00**	99.92±0.07
14	97.11±2.49	86.56±10.95	93.89±7.56	83.44±13.94	90.86±6.92	98.12±1.43	91.11±3.89	77.78±9.12	**99.78±0.45**	94.67±10.2
OA(%)	80.55±1.21	89.82±1.01	81.3±1.56	86.59±1.39	89.07±1.27	95.40±1.32	94.46±2.67	97.08±0.14	96.54±0.85	**98.56±0.13**
AA(%)	81.43±1.30	90.1±1.21	81.28±1.82	86.72±1.53	89.15±1.27	94.70±1.56	92.48±1.34	96.19±1.21	96.3±0.81	**98.29±0.23**
k×100	78.93±1.31	88.97±1.10	79.73±1.70	85.47±1.50	88.15±1.37	95.01±1.13	93.99±0.91	96.84±1.01	96.25±0.92	**98.44±0.15**

TABLE IX

CLASSIFICATION PERFORMANCE OF THE XUZHOU DATASET OBTAINED BY DIFFERENT METHODS (THE BEST RESULTS ARE SHOWN IN BOLD)

No.	Random Forest [52]	MLR [53]	SVM [20]	MLP [54]	CNN [55]	HybridSN [29]	DW-KELM [56]	MSC-EDKELM [47]	MSAF-KELM
1	92.81±1.37	93.06±1.00	93.7±1.25	93.48±0.72	94.85±1.05	94.40±0.82	**97.43±0.34**	95.87±2.12	96.74±0.91
2	94.83±1.85	97.59±0.14	96.71±0.32	97.43±0.29	98.12±0.40	96.17±1.86	97.88±0.11	**99.40±0.01**	98.63±0.54
3	69.15±9.83	82.65±4.37	74.41±7.79	85.12±5.80	84.77±7.25	91.06±5.53	91.69±0.42	**96.82±2.03**	92.22±5.38
4	78.41±5.88	87.74±3.57	84.69±3.54	89.50±3.68	**92.82±1.94**	86.43±8.19	81.32±3.28	83.40±4.32	84.65±5.67
5	89.15±2.21	93.17±1.82	94.21±2.06	91.62±2.13	95.84±1.26	95.82±2.07	89.08±3.11	**96.16±1.39**	94.95±4.26
6	49.44±6.68	60.64±4.36	39.79±10.27	59.05±5.35	65.89±6.19	81.17±13.09	92.12±2.46	92.99±2.34	**93.16±4.98**
7	73.54±4.30	77.72±3.68	62.16±5.65	78.12±1.80	82.44±2.92	97.44±0.87	94.39±2.56	**98.46±0.03**	97.31±3.01
8	96.13±0.53	96.32±0.83	95.74±0.66	95.11±1.46	96.97±1.25	**99.40±0.40**	96.49±0.54	94.57±2.15	95.56±2.18
9	78.88±9.42	92.56±3.43	72.82±6.83	93.24±3.26	**98.14±0.53**	96.52±1.40	79.90±3.28	77.94±9.53	84.11±7.51
OA(%)	86.3±1.06	90.02±0.42	86.61±0.63	90.04±0.52	92.68±0.76	94.32±1.26	93.06±0.45	94.50±0.22	**94.70±0.12**
AA(%)	80.26±1.82	86.83±0.77	79.36±1.69	86.96±0.77	89.98±1.30	92.16±1.60	91.15±9.21	92.85±0.11	**93.04±0.55**
k×100	82.49±1.36	87.31±0.56	82.77±0.83	87.35±0.64	90.68±0.97	92.82±1.62	91.14±0.32	93.02±0.13	**93.25±0.20**

However, due to the low classification progress of category 9 (i.e., Shadows), the AA value of the MSAF-KELM was lower than that of SSFTT. Taking all data indicators together, our method was still highly competitive.

Table VIII shows the experimental performance of the Botswana dataset, a large spatial scale dataset. Due to the spatial complexity, the spatial size reaches 1476 × 256, the types of features are complex, and the number of samples is small. Therefore, a sampling rate of 5% was chosen as the training set. In this dataset, we exploited the efficient classification of ultrasmall sample rates to the extreme on complex topographic surfaces because our method fully extracted the spectral and spatial features at different scales. For the MSC-EDKELM classifier, due to the rich context information of multiple scales, it had also achieved good classification accuracy, but it was still lower than our method. The performances of the DW-KELM, SSFTT, and HybridSN methods were slightly lower than that of MSC-EDKELM.

Finally, the Xuzhou dataset results are shown in Table IX. Xuzhou is the dataset with the highest spatial resolution in our experiments, reaching 0.73 m/pixel. This means that more detailed information is processed when dealing with this HSI data, which is a challenge for all classifiers. An ultrasmall sample rate of 0.5% was chosen as the training set. For this dataset, the classification performance of the deep learning methods was obviously better than that of traditional methods. Among them, the classification accuracies of HybridSN and DW-KELM were close to the MS MSC-EDKELM method.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:54:29 UTC from IEEE Xplore. Restrictions apply.

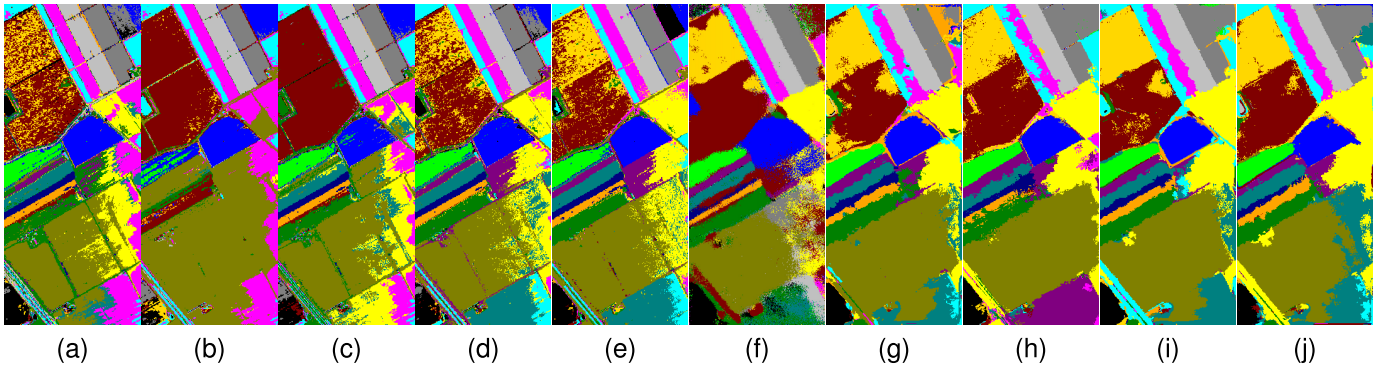


Fig. 12. Classification maps for the Salinas database under different algorithms. (a) RF. (b) MLR. (c) SVM. (d) MLP. (e) CNN. (f) HybridSN. (g) DW-KELM. (h) MSC-EDKELM. (i) SSFTT. (j) MSAF-KELM.

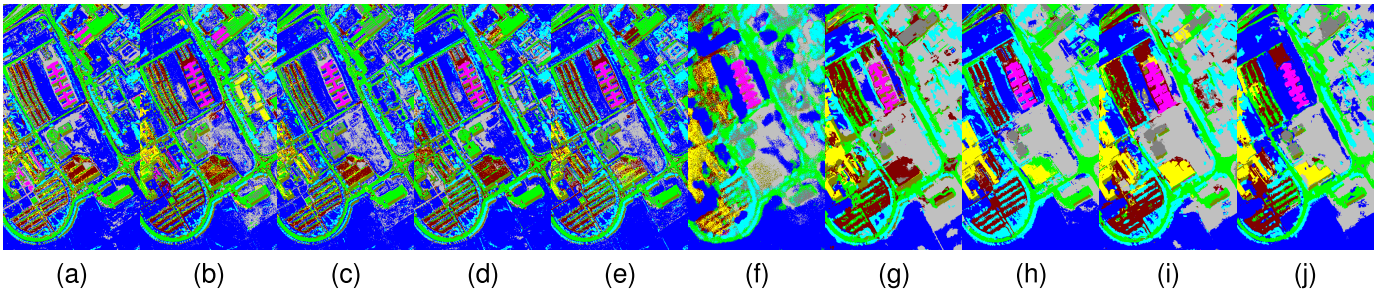


Fig. 13. Classification maps for the Pavia University database under different algorithms. (a) RF. (b) MLR. (c) SVM. (d) MLP. (e) CNN. (f) HybridSN. (g) DW-KELM. (h) MSC-EDKELM. (i) SSFTT. (j) MSAF-KELM.

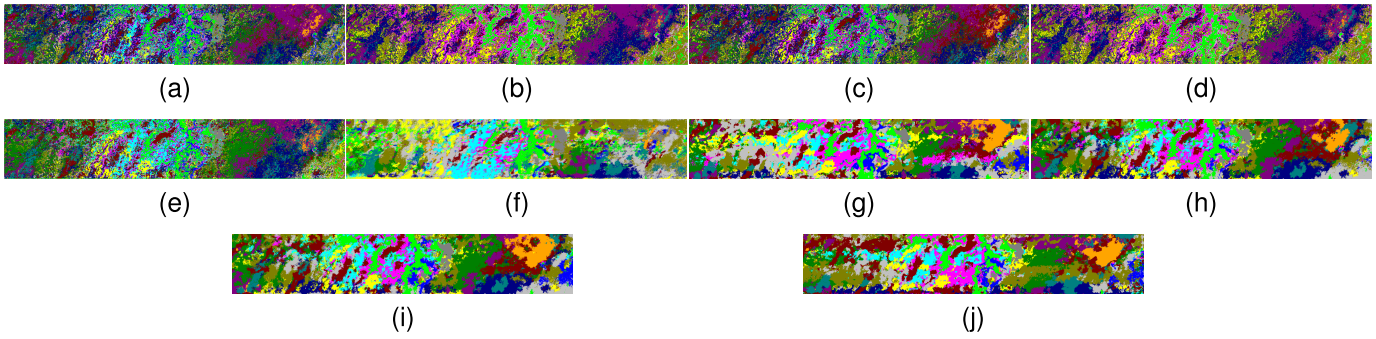


Fig. 14. Classification maps for the Botswana database under different algorithms. (a) RF. (b) MLR. (c) SVM. (d) MLP. (e) CNN. (f) HybridSN. (g) DW-KELM. (h) MSC-EDKELM. (i) SSFTT. (j) MSAF-KELM.

Benefitting from the rich MS features, the classification accuracy of each category was relatively stable for our method. Consequently, our method, MSAF-KELM, achieved the highest accuracies among all comparison methods.

E. Visual Evaluation

The classification maps of the proposed MSAF-KELM and several compared methods are shown in Figs. 12–15. The classification experiments were run on four datasets: Salinas, Pavia University, Botswana, and Xuzhou. By visual comparison, the classification maps obtained by MSAF-KELM methods were the cleanest and the most accurate, and the closest to the ground-truth maps. Traditional machine learning algorithms, such as RF, MLR, and SVM, contain obvious noises in the classification maps. Deep learning methods, MLP and CNN, have reduced the noise phenomenon, but there are still some obvious areas of misclassification. For methods such as HybridSN, DW-KELM, SSFTT, and MSC-EDKELM, the accuracies are much higher than those of the machine learning-based methods, but the contours of the category labels are not rounded enough, and the label boundaries are not clear enough. In summary, our method is competitive in terms of noise phenomena, classification accuracy, roundness, and sharpness of classification label contours.

Specifically, for the Salinas dataset, the comparison reveals that all three traditional algorithms incorrectly classify the *Vineyard_untrained* category (in yellow) in the upper left corner as a *Grapes_untrained* category (in red-brown). Deep learning methods contain obvious noise effects in the *Vineyard_untrained* category (in red-brown), and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:54:29 UTC from IEEE Xplore. Restrictions apply.

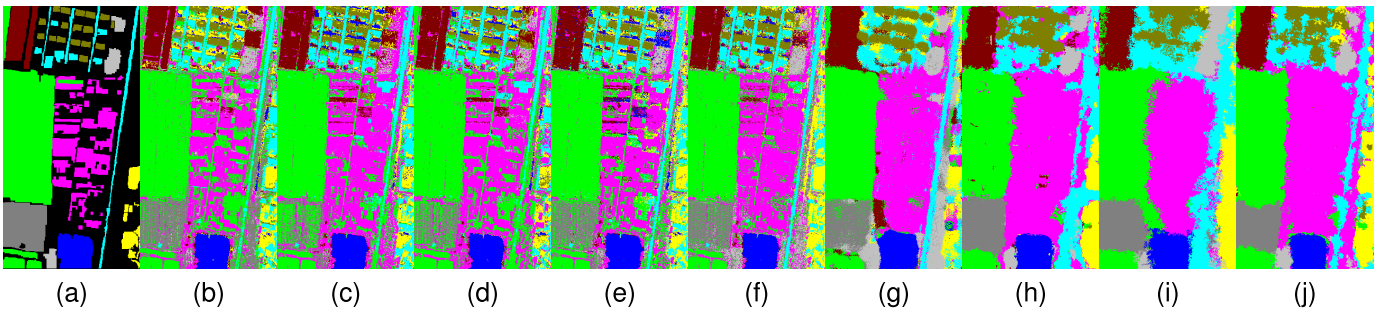


Fig. 15. Classification maps for the Xuzhou database under different algorithms. (a) Ground-truth map. (b) RF. (c) MLR. (d) SVM. (e) MLP. (f) CNN. (g) HybridSN. (h) DW-KELM. (i) MSC-EDKELM. (j) MSAF-KELM.

the HybridSN and SSFTT methods also have misclassification in the Vineyard untrained category. DW-KELM, MSC-EDKELM, and MSAF-KELM methods can classify both categories, but MSAF-KELM has fewer noises than others. This validates the superior performance of our method. Second, for the Pavia University dataset, the RF, MLR, SVM, MLP, and CNN methods have more severe salt-and-pepper noise phenomena in their classification maps. The contours between the category labels of the HybridSN, DW-KELM, MSC-EDKELM, and SSFTT methods are not clear. By comparing several boundary regions, our method maintains the best boundary classification performance. Third, for the Botswana dataset, our method has fewer noise points and can maintain clear bounds. The proposed method also exhibits good classification performance for complex large-scale datasets. Finally, for the Xuzhou dataset, for the middle Crop 1 (in red) and Crop 2 (in green) categories, the boundary contours of the other comparison methods are less clear and noisier. Our method is not only the closest to the ground truth but also has the clearest boundary.

In conclusion, the visual comparison of several methods further demonstrates the excellent classification performance of our method.

F. Comparison of Fusion Methods

In our proposed MSAF-KELM method, the most important step in determining the accuracy of the final classification results is the executing WASFS module. Our proposed WSAFS module considers the classification accuracy of each structure in the training and testing phases while adaptively using the self-attention mechanism as one of the weights for fusion selection. To validate the fusion performance of WSAFS, we compare it with two common fusion methods: the majority voting strategy (MVS) and WOLFS. We conducted comparative experiments using the Salinas and Pavia University datasets with 15 and 30 labels on each category as training sets, respectively. The experimental results are shown in Table X.

As seen in Table X, the MVS only considers the number of pixel classification labels in different structures and converts the classification results into a probability problem by voting to obtain the labels with the maximum probability, which lacks specific analysis of individual feature and category classification. The WOLFS considers the classification accuracy of each structure during training and testing, and uses them as voting weights to obtain the maximum probability labels. However, it lacks the consideration of the correlation of individual features. The WSAFS fusion strategy not only considers the training and testing accuracy but also cleverly combines the self-attention mechanism weights to form joint weights to determine the final classification results. The result of WSAFS is the best among the two fusion strategies.

TABLE X
COMPARISON OF CLASSIFICATION RESULTS USING DIFFERENT FUSION METHODS UNDER SALINAS AND PAVIA UNIVERSITY DATASETS (THE BEST RESULTS ARE SHOWN IN BOLD)

Fusion Method	MVS	WOLFS	WSAFS
	Salinas		
OA(%)	93.56	94.81	**96.54**
AA(%)	96.26	95.73	**97.11**
k*100	92.85	94.23	**96.15**
	Pavia University		
OA(%)	93.01	95.86	**96.44**
AA(%)	93.89	93.09	**93.34**
k*100	90.90	94.50	**95.27**

G. Comparison of the Influence of Training Sample Size

To further evaluate the generalization ability of MSAF-KELM in different settings, we performed comparison experiments with CNN, HybridSN, MSC-EDKELM, SSFTT, and DW-KELM for training sample size. We set the number of samples for each category to 3, 6, 9, 12, and 15 on the Salinas, Pavia University, Botswana, and Xuzhou datasets, respectively. The experimental results are shown in Fig. 16. It is obvious that the classification accuracy is proportional to the number of samples, and MSAF-KELM always maintains the highest accuracy. At the same time, the proposed method can maintain high accuracy even with a sample size of only 3 for each category, which demonstrates that MSAF-KELM can extract spectral–spatial features in depth for ultrasmall rates. In addition, thanks to the ability of WSAFS to exploit the advantages of each scale feature, it enables MSAF-KELM to capture different feature information at small sample size.

H. Comparison of Time Cost

We compared the training and testing time of the proposed method with CNN, HybridSN, MSC-EDKELM, SSFTT, and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:54:29 UTC from IEEE Xplore. Restrictions apply.

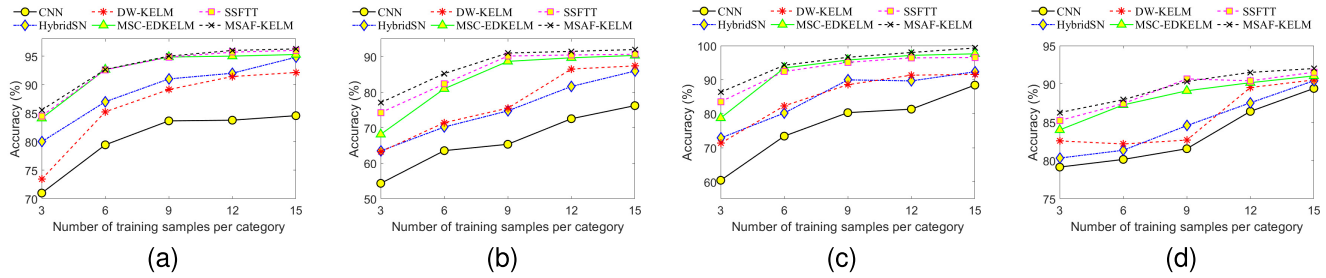


Fig. 16. Influence of different training sample sizes of CNN, HybridSN, MSC-EDKELM, SSFTT, and DW-KELM algorithms on classification accuracy. (a) Salinas dataset. (b) Pavia University dataset. (c) Botswana dataset. (d) Xuzhou dataset.

TABLE XI

TRAINING TIME AND TEST TIME IN SECONDS (S) BETWEEN THE CONTRAST METHODS AND THE PROPOSED METHOD ON FOUR DATASETS (THE BEST RESULTS ARE SHOWN IN BOLD)

Methods	Salinas		Pavia University		Botswana		Xuzhou	
	Train (s)	Test (s)	Train (s)	Test (s)	Train (s)	Test (s)	Train (s)	Test (s)
CNN [55]	11.26	6.54	5.13	3.21	7.28	2.79	12.42	2.42
HybridSN [29]	380.27	37.61	386.64	17.91	372.26	4.45	375.54	20.64
MSC-EDKELM [47]	0.13	4.32	0.07	2.82	**0.03**	**0.16**	**0.75**	8.86
SSFTT [57]	24.44	15.06	12.54	8.45	16.12	7.87	28.11	17.42
DW-KELM [56]	**0.10**	**2.46**	**0.024**	**1.98**	**0.03**	0.27	0.79	**5.97**
MSAF-KELM	0.39	6.58	0.04	3.44	**0.03**	0.34	0.82	8.82

DW-KELM. The numbers of training and testing samples are distributed in Tables I–IV, and the comparison results are shown in Table XI. It is obvious that MSAF-KELM is relatively fast, and KELM is more suitable for MS and multifeature classification tasks than deep learning methods due to its simple and efficient network structure. MSC-EDKELM uses a smaller number of features than our proposed method, and the efficient WSAFS mechanism also has some time consumption. Thus, it is slightly faster than our method. For DW-KELM, it is the fastest because of its simple structure and uses only a single scale to construct the spectral–spatial kernel. Thus, its classification accuracy is lower than our proposed method. Combining the classification accuracies from Tables VI–IX, it can be seen that MSAF-KELM achieves the optimal classification effect in a shorter time.

IV. CONCLUSION

In this article, a novel MSAF-KELM is proposed to achieve effective HSI classification at ultrasmall sample rates. The method uses a multi-structured feature extraction module to exploit information on small sample size. Then, the MS features are input to multiple KELM classifiers for fast classification. In addition, WSAFS is used for accurate and efficient fusion, which allows the limited sample information to be utilized to the maximum extent. The experimental results of this method on four datasets also demonstrate the excellent classification performance of the proposed method at ultrasmall sample rates.

In the future, we will embed MS and multifeature methods into the transformer network to further improve the classification performance at ultrasmall sample rates. The semantic information of each category will also be incorporated into the network to enhance the processing of label boundary information and improve the overall classification accuracy.

REFERENCES

[1] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. M. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[2] L. He, J. Li, C. Liu, and S. Li, "Recent advances on spectral–spatial hyperspectral image classification: An overview and new guidelines," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1579–1597, Mar. 2017.

[3] C. He, L. Sun, W. Huang, J. Zhang, Y. Zheng, and B. Jeon, "TSLRLN: Tensor subspace low-rank learning with non-local prior for hyperspectral image mixed denoising," *Signal Process.*, vol. 184, 2021, Art. no. 108060.

[4] L. Sun, C. He, Y. Zheng, and S. Tang, "SLRL4D: Joint restoration of subspace low-rank learning and non-local 4-D transform filtering for hyperspectral image," *Remote Sens.*, vol. 12, no. 18, p. 2979, 2020.

[5] Y. Kong, X. Wang, and Y. Cheng, "Spectral–spatial feature extraction for HSI classification based on supervised hypergraph and sample expanded CNN," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 11, pp. 4128–4140, Nov. 2018.

[6] F. Luo, Z. Zou, J. Liu, and Z. Lin, "Dimensionality reduction and classification of hyperspectral image via multistructure unified discriminative embedding," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2021.

[7] D. Hong *et al.*, "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[8] D. Hong, N. Yokoya, G.-S. Xia, J. Chanussot, and X. X. Zhu, "X-ModalNet: A semi-supervised deep cross-modal network for classification of remote sensing data," *ISPRS J. Photogramm. Remote Sens.*, vol. 167, pp. 12–23, Sep. 2020.

[9] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2018.

[10] X. Liu, D. Hong, J. Chanussot, B. Zhao, and P. Ghamisi, "Modality translation in remote sensing time series," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2021.

[11] J. Wang, L. Zhang, Q. Tong, and X. Sun, "The spectral crust project—Research on new mineral exploration technology," in *Proc. 4th Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Jun. 2012, pp. 1–4.

[12] A. Fong, G. Shu, and B. McDonogh, "Farm to table: Applications for new hyperspectral imaging technologies in precision agriculture, food quality and safety," in *Proc. CLEO, Appl. Technol.* Washington, DC, USA: Optical Society of America, 2020, p. AW3K-2.

[13] M. Lv, W. Li, T. Chen, J. Zhou, and R. Tao, "Discriminant tensor-based manifold embedding for medical hyperspectral imagery," *IEEE J. Biomed. Health Informat.*, vol. 25, no. 9, pp. 3517–3528, Sep. 2021.

[14] F. Tong and Y. Zhang, "Spectral–spatial and cascaded multilayer random forests for tree species classification in airborne hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022.

[15] H. Su, Y. Yu, Z. Wu, and Q. Du, "Random subspace-based k-nearest class collaborative representation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6840–6853, Aug. 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:54:29 UTC from IEEE Xplore. Restrictions apply.

[16] Y. Chen *et al.*, "Unsupervised Bayesian subpixel mapping of hyperspectral imagery based on band-weighted discrete spectral mixture model and Markov random field," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 1, pp. 162–166, Jan. 2020.

[17] M. Sheykhmousa, M. Mahdianpari, H. Ghanbari, F. Mohammadimanesh, P. Ghamisi, and S. Homayouni, "Support vector machine versus random forest for remote sensing image classification: A meta-analysis and systematic review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 6308–6325, 2020.

[18] M. Ahmad *et al.*, "Hyperspectral image classification—Traditional to deep models: A survey for future prospects," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 968–999, 2022.

[19] Q. Hao, S. Li, L. Fang, and X. Kang, "Multiscale feature extraction with Gaussian curvature filter for hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Oct. 2020, pp. 80–83.

[20] M. Fauvel, J. A. Benediktsson, J. Chanussot, and J. R. Sveinsson, "Spectral and spatial classification of hyperspectral data using SVMs and morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 11, pp. 3804–3814, Nov. 2008.

[21] J. A. Benediktsson, J. A. Palmason, and J. R. Sveinsson, "Classification of hyperspectral data from urban areas based on extended morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 480–491, Mar. 2005.

[22] M. M. Dalla, A. Villa, J. A. Benediktsson, J. Chanussot, and L. Bruzzone, "Classification of hyperspectral images by using extended morphological attribute profiles and independent component analysis," *IEEE Geosci. Remote Sens. Lett.* vol. 8, no. 3, pp. 542–546, Dec. 2010.

[23] L. Sun, F. Wu, T. Zhan, W. Liu, J. Wang, and B. Jeon, "Weighted nonlocal low-rank tensor decomposition method for sparse unmixing of hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1174–1188, 2020.

[24] L. Fu *et al.*, "Learning robust discriminant subspace based on joint $L_{2,p}$- and $L_{2,s}$-norm distance metrics," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 1, pp. 130–144, Jan. 2022.

[25] Q. Ye, Z. Li, L. Fu, Z. Zhang, W. Yang, and G. Yang, "Nonpeaked discriminant analysis for data representation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 12, pp. 3818–3832, Dec. 2019.

[26] Q. Ye, P. Huang, Z. Zhang, Y. Zheng, L. Fu, and W. Yang, "Multiview learning with robust double-sided twin SVM," *IEEE Trans. Cybern.*, early access, Sep. 21, 2021, doi: 10.1109/TCYB.2021.3088519.

[27] Q. Yan, J. Zhang, and J. Feng, "Spectral–spatial classification of hyperspectral image using PCA and Gabor filtering," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Oct. 2020, pp. 513–516.

[28] W. Li, G. Wu, F. Zhang, and Q. Du, "Hyperspectral image classification using deep pixel-pair features," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 844–853, Feb. 2016.

[29] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[30] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2020.

[31] S. Wan, S. Pan, P. Zhong, X. Chang, J. Yang, and C. Gong, "Dual interactive graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[32] D. Hong *et al.*, "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, Aug. 2020.

[33] D. Hong, J. Hu, J. Yao, J. Chanussot, and X. X. Zhu, "Multimodal remote sensing benchmark datasets for land cover classification with a shared and specific feature learning model," *ISPRS J. Photogramm. Remote Sens.*, vol. 178, pp. 68–80, Aug. 2021.

[34] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–10, 2022.

[35] M. S. Kotzagiannidis and C.-B. Schönlieb, "Semi-supervised superpixel-based multi-feature graph learning for hyperspectral image data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022.

[36] B. Yin and B. Cui, "Multi-feature extraction method based on Gaussian pyramid and weighted voting for hyperspectral image classification," in *Proc. IEEE Int. Conf. Consum. Electron. Comput. Eng. (ICCECE)*, Jan. 2021, pp. 645–648.

[37] P. Duan, X. Kang, S. Li, and P. Ghamisi, "Noise-robust hyperspectral image classification via multi-scale total variation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 6, pp. 1948–1962, Jun. 2019.

[38] L. Sun, C. Ma, H. J. Shim, Z. Wu, and B. Jeon, "Adjacent superpixel-based multiscale spatial–spectral kernel for hyperspectral classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 6, pp. 1905–1919, Jun. 2019.

[39] W. Huang, Y. Huang, H. Wang, Y. Liu, and H. Jae Shim, "Local binary patterns and superpixel-based multiple kernels for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4550–4563, 2020.

[40] S. Jia, L. Shen, J. Zhu, and Q. Li, "A 3-D Gabor phase-based coding and matching framework for hyperspectral imagery classification," *IEEE Trans. Cybern.*, vol. 48, no. 4, pp. 1176–1188, Apr. 2017.

[41] H. Chen, F. Miao, Y. Chen, Y. Xiong, and T. Chen, "A hyperspectral image classification method using multifeature vectors and optimized KELM," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2781–2795, 2021.

[42] X. Liu, Q. Hu, Y. Cai, and Z. Cai, "Extreme learning machine-based ensemble transfer learning for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3892–3902, 2020.

[43] F. Cao, Z. Yang, J. Ren, W. Chen, G. Han, and Y. Shen, "Local block multilayer sparse extreme learning machine for effective feature extraction and classification of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5580–5594, Aug. 2019.

[44] J. Li, B. Xi, Q. Du, R. Song, Y. Li, and G. Ren, "Deep kernel extreme-learning machine for the spectral–spatial classification of hyperspectral imagery," *Remote Sens.*, vol. 10, no. 12, p. 2036, Dec. 2018.

[45] Y. Liang, L. Wei, and Q. Lu, "Multiple feature fusion for fine classification of crops in UAV hyperspectral imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 5059–5062.

[46] C. Zhang, G. Li, and S. Du, "Multi-scale dense networks for hyperspectral remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9201–9222, Aug. 2019.

[47] B. Xi, J. Li, Y. Li, R. Song, W. Sun, and Q. Du, "Multiscale context-aware ensemble deep KELM for efficient hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5114–5130, Jun. 2020.

[48] C. Pu, H. Huang, and L. Luo, "Classfication of hyperspectral image with attention mechanism-based dual-path convolutional network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[49] Y. Xu, B. Du, and L. Zhang, "Self-attention context network: Addressing the threat of adversarial attacks for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 30, pp. 8671–8685, 2021.

[50] S. Jia, X. Deng, J. Zhu, M. Xu, J. Zhou, and X. Jia, "Collaborative representation-based multiscale superpixel fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7770–7784, Oct. 2019.

[51] Y. Tang, L. Zhao, and L. Ren, "Different versions of entropy rate superpixel segmentation for hyperspectral image," in *Proc. IEEE 4th Int. Conf. Signal Image Process. (ICSIP)*, Jul. 2019, pp. 1050–1054.

[52] J. Ham, Y. Chen, M. M. Crawford, and J. Ghosh, "Investigation of the random forest framework for classification of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 492–501, Mar. 2005.

[53] J. Li, J. Bioucas-Dias, and A. Plaza, "Spectral-spatial hyperspectral image segmentation using subspace multinomial logistic regression and Markov random fields," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 3, pp. 809–823, 2011.

[54] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Deep learning classifiers for hyperspectral imaging: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 279–317, Dec. 2019.

[55] K. Makantasis, K. Karantzalos, A. Doulamis, and N. Doulamis, "Deep supervised learning for hyperspectral data classification through convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4959–4962.

[56] X. Yu, Y. Feng, Y. Gao, Y. Jia, and S. Mei, "Dual-weighted kernel extreme learning machine for hyperspectral imagery classification," *Remote Sens.*, vol. 13, no. 3, p. 508, 2021.

[57] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:54:29 UTC from IEEE Xplore. Restrictions apply.



Le Sun (Member, IEEE) was born in Jiangsu, China, in 1987. He received the B.S. degree from the School of Science, Nanjing University of Science and Technology (NJUST), Nanjing, Jiangsu, in 2009, and the Ph.D. degree from the School of Computer Science and Engineering, NJUST, in 2014.

From 2015 to 2018, he conducted research in the field of multi-images' fusion based on sparse dictionary learning and compressive sensing as a Post-Doctoral Researcher at the School of Electronic and Electrical Engineering, Sungkyunkwan University, Seoul, South Korea. Since 2020, he has been an Associate Professor with the School of Computer and Science, Nanjing University of Information Science and Technology, Nanjing. His research interests include hyperspectral image processing (including unmixing, classification, and restoration), sparse representation, compressive sensing, and deep learning.



Yu Fang was born in Jiangsu, China, in 1998. He is currently pursuing the M.S. degree in electronic information with the School of Computer and Science, Nanjing University of Information Science and Technology, Nanjing, China, in 2021.

His research interests include hyperspectral image processing.



Yuwen Chen received the master's degree from the University of Science and Technology of China, Hefei, China, in 2011.

He is currently an Associate Researcher with the Chongqing Institute of Green and Intelligent Technology, Chinese Academy of Sciences, Chongqing, China. He has long conducted research on video semantic cognition and intelligent software for big data analysis.



Wei Huang (Member, IEEE) was born in Henan, China, in 1982. He received the Ph.D. degree in pattern recognition and intelligence systems from the Nanjing University of Science and Technology (NUST), Nanjing, China, in 2015.

He is currently an Associate Professor with the School of Computer and Communication Engineering, Zhengzhou University of Light Industry, Zhengzhou, China. His research interests include hyperspectral classification and image processing.



Zebin Wu (Senior Member, IEEE) was born in Zhejiang, China, in 1981. He received the B.S. and Ph.D. degrees from the School of Computer Science and Engineering, Nanjing University of Science and Technology (NJUST), Nanjing, Jiangsu, China, in 2003 and 2008, respectively.

He is currently a Full Professor with the School of Computer Science and Engineering, NJUST. His research interests include virtual reality and system simulation, remote sensing information processing, and distributed computing.



Byeungwoo Jeon (Senior Member, IEEE) received the B.S. degree *(magna cum laude)* in electronics engineering and the M.S. degree in electronics engineering from Seoul National University, Seoul, South Korea, in 1985 and 1987, respectively, and the Ph.D. degree in electrical engineering from Purdue University, West Lafayette, IN, USA, in 1992.

He is currently a Professor with the School of Electronic and Electrical Engineering, Sungkyunkwan University, Seoul. His main research areas include multimedia signal processing, video compression, statistical pattern recognition, and remote sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:54:29 UTC from IEEE Xplore. Restrictions apply.